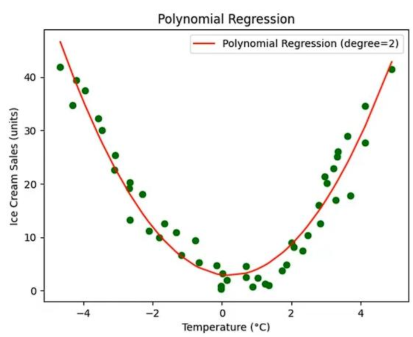
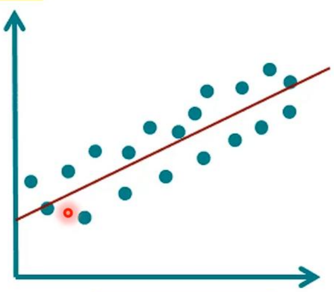
08_Coefficient of Determination

The **coefficient of determination**, often denoted as **R^2 (R-squared)**, is a statistical measure used to assess how well a regression model explains the variability of the dependent variable.

It tells you **how much of the variation in the output (Y)** can be **explained by the input features (X)** in a regression model. R^2 measures how well **any regression model,** including linear, polynomial, and neural network, explains the **variance in the target variable**, with 1 being a perfect fit. Range: $-\infty < R^2 \leq 1$





Formula: $$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

- SS_{res} = Sum of Squares of Residuals
- SS_{tot} = Total Sum of Squares

Interpretation:

- **R^2 = 1**: Perfect fit. The model explains 100% of the variance.
- **R^2 = 0**: The model explains none of the variance—no better than just using the mean.
- **R^2 < 0**: Model is worse than just predicting the mean (can happen with bad models).

$$\mathrm{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$